1.1 Functions and Their Representations

Calculus 1

Sang June Lee

Kyung Hee University, Korea

Textbook: James Stewart, Essential Calculus - Early Transcendentals

Outline

< Part 1 >

Basics of Functions

Motivation

- Functions are defined whenever one quantity depends on another.
- **Examples:**

(1) The area A of a circle depends on the radius r of the circle.

  - We know that $A = \pi r^2$.
  - We say that A is a **function** of r.

(2) The human population P of the world depends on the time t.

  - Let $P(t)$ be the world population at time t.

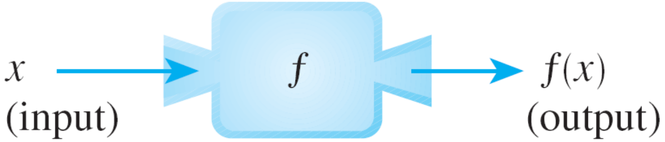
Function

Definition: A **function** f is a rule that assigns to each element x in a set D exactly one element, called $f(x)$, in a set E.

- D and E are sets of real numbers.
- D is called the **domain** of the function.
- $f(x)$is the **value of** f **at** x and is read "f of$x$".
- The **range** E of f is the set of all possible values of $f(x)$ as x varies in D.
- An number in the domain of a function f is called an **independent variable**.
- A number in the range of f is called a **dependent variable**.

Visualization 1: Machine diagram

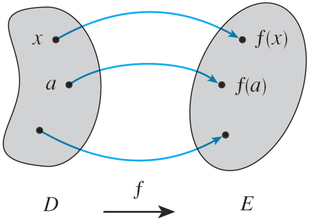
It's helpful to think of a function as a **machine**.



Visualization 2: Arrow diagram

Another way to picture a function is by an **arrow diagram**.

- Each arrow connects an element of D to an element of E.
- The arrow indicates that $f(x)$ is associated with x.

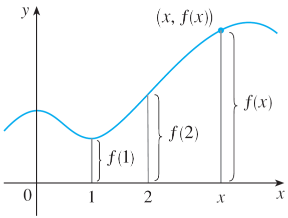


Visualization 3: Graph

The most common method for visualizing a function.

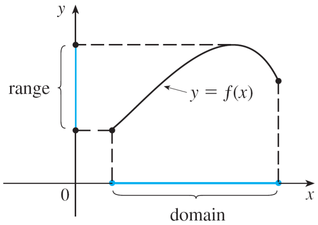
If f is a function with domain D, then its **graph** is the set of ordered pairs $(x, f(x))$.

$f(x) =$ height of the graph above the point x.



Domain and range in a graph

The graph of f also allows us to picture the **domain** of f on the x-axis and its **range** on the y-axis.

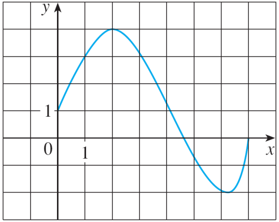


Example 1

The graph of a function f is shown below.

(a) Find the values of $f(1)$ and $f(5)$.

(b) What are the domain and range of f?



Example 1 - Solution

(a) $(1, 3)$ lies on the graph of f, so $f(1) = 3$.

When $x = 5$, the graph lies about 0.7 unit below the x-axis, so, we estimate that $f(5) \approx -0.7$.

(b) $f(x)$ is defined when $0 \leq x \leq 7$, so, the domain of f is $[0, 7]$.

Notice that f takes on all values from -2 to 4, so, the range of f is $[-2, 4]$.

Note

< Part 2 >
Representations of Functions

4 Representations of Functions

There are four possible ways to represent a function:

1. Verbally (by a description in words)
2. Visually (by a graph)
3. Numerically (by a table of values)
4. Algebraically (by an explicit formula)

Remark

If a single function can be represented in all four ways, it is often useful to go from one representation to another to gain additional insight into the function.

But certain functions are described more naturally by one method than by another.

Example 1

The most useful representation of the area of a circle as a function of its radius is probably the algebraic formula

$$A(r) = \pi r^2,$$

though it is possible to compile a table of values or to sketch a graph (half a parabola).

Because a circle has a positive radius, the domain is $(0, \infty)$, and the range is also $(0, \infty)$.

Example 2

We are given a description of the function in words:

$P(t)$ is the human population of the world at time t.

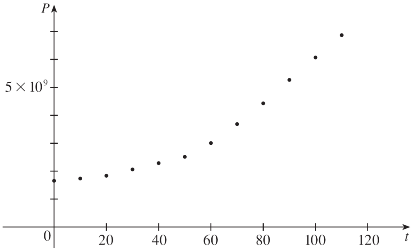
$t = 0$ corresponds to the year 1900.

Population	t (millions)
0	1650
10	1750
20	1860
30	2070
40	2300
50	2560
60	3040
70	3710

Graph

If we plot these values, we get the graph (called a **scatter plot**).

Scatter plot of data points for population growth

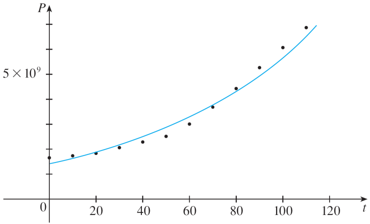


Mathematical model

It is possible to find an expression for a function that **approximates** $P(t)$, we could use a graphing calculator with exponential regression capabilities to obtain the approximation.

$$P(t) \approx f(t) = (1.43653 \times 10^9)(1.01395)^t.$$

The function f is called a **mathematical model**for population growth.



Example 2

When you turn on a hot-water faucet, the temperature T of the water depends on how long the water has been running.

Draw a rough graph of T as a function of the time t that has elapsed since the faucet was turned on.

Solution: (Next page!)

Example 2 - Solution

The initial temperature = room temperature

When the water from the hot water tank starts flowing from the faucet, T increases quickly.

Next, T is constant at the temperature of the heated water in the tank.

When the tank is drained, Tdecreases to the temperature of the water.

Sketch of T as a function of t :

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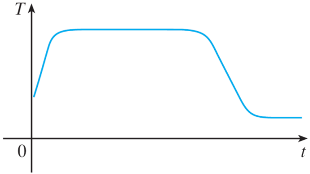


Vertical line test

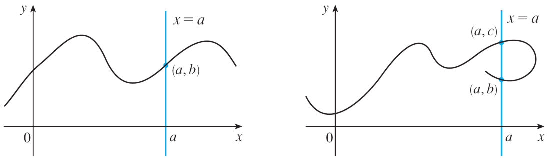
The graph of a function is a curve in the *xy*-plane.

Question: Which curves in the *xy*-plane are graphs of functions?

Answer:

Vertical line test

A curve in the *xy*-plane is the graph of a function of x if and only if no vertical line intersects the curve more than once.



Note

< Part 3 >

Piecewise Defined Functions

Piecewise Defined Functions

The **absolute value** of a number a, denoted by $|a|$, is the distance from a to 0 on the real number line.

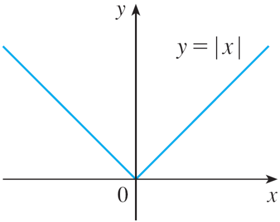
- $|a| \geq 0$ for every number a
- For example,
  $|3| = 3, |-3| = 3, |0| = 0, |\sqrt{2} - 1| = \sqrt{2} - 1, |3 - \pi| = \pi - 3$
- In general, $|a| = a$ if $a \geq 0$
  $|a| = -a$ if $a < 0$

Example 5

Sketch the graph of the absolute value function $f(x) = |x|$.

Solution:

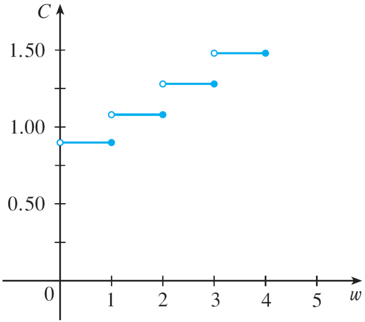
$$f(x) = |x| = \begin{cases} x, & x \geq 0, \\ -x, & x < 0. \end{cases}$$



Example 6

$C(w)=$ a cost of mailing a large envelope with weight w.

$$C(w) = \begin{cases} 0.88 & \text{if } 0 < w \leqslant 1 \\ 1.08 & \text{if } 1 < w \leqslant 2 \\ 1.28 & \text{if } 2 < w \leqslant 3 \\ 1.48 & \text{if } 3 < w \leqslant 4 \end{cases}$$



Note

< Part 4 >

Symmetry

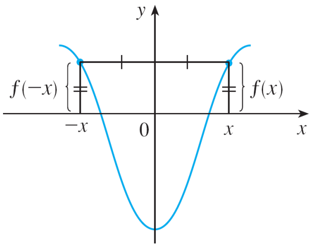
Even function

If $f(-x) = f(x)$,then f is called an **even function**.

For instance, the function $f(x) = x^2$ is even because

$$f(-x) = (-x)^2 = x^2 = f(x).$$

Its graph is symmetric with respect to the y-axis:

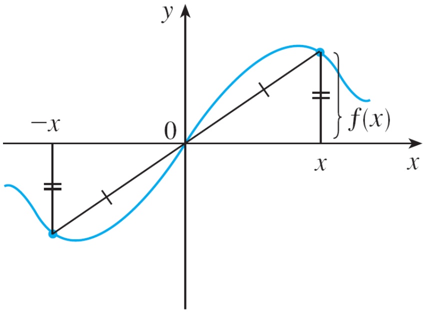


Odd function

If $f(-x) = -f(x)$, then f is called an **odd function**.

For example, the function $f(x) = x^3$is odd because

The graph of an odd function is symmetric about the origin.



Example 7

Determine whether each of the following functions is even, odd, or neither even nor odd.

(a) $f(x) = x^5 + x$

(b) $g(x) = 1 - x^4$

(c) $h(x) = 2x - x^2$

Example 7 - Solution

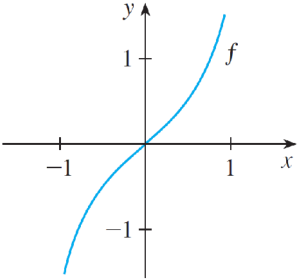
(a) $f(-x) = (-x)^5 + (-x) = (-1)^5x^5 + (-x) = -x^5 - x = -(x^5 + x) = -f(x)$
Therefore, f is an odd function.

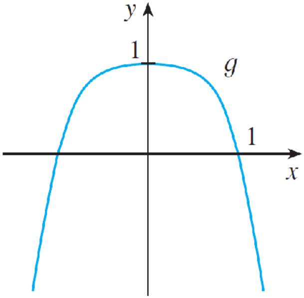
(b) $g(-x) = 1 - (-x)^4 = 1 - x^4 = g(x)$
So, g is even.

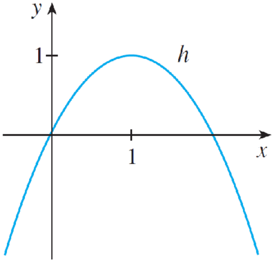
(c) $h(-x) = 2(-x) - (-x)^2 = -2x - x^2$
Since $h(-x) \neq h(x)$and $h(-x) \neq -h(x)$, the function $h$is neither even nor odd.

Example 7 - graphs

The graphs of the functions in Example 7 are shown.







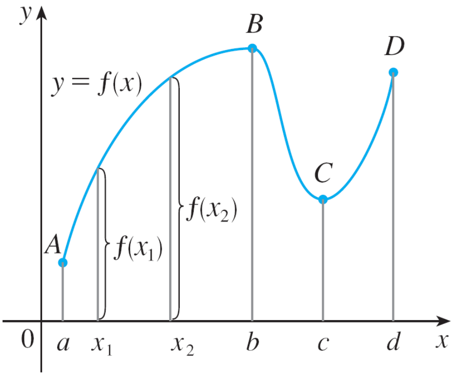
Note

< Part 5 >

Increasing and Decreasing Functions

Increasing and Decreasing Functions

The function f is said to be **increasing** on the interval $[a, b]$, **decreasing** on $[b, c]$, and **increasing** again on $[c, d]$.



Formal definition

- A function f is called **increasing** on an interval I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ in I.
- A function f is called **decreasing** on an interval I if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ in I.

Note